MILITARY SPECIFICATION

GYROSCOPE, RATE SWITCHING CN-495/AJB-3

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design and all performance requirements for the rate switching gyroscope CN-495/AJB-3.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents, of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen
QQ-C-320	Chromium Plating (Electrodeposited)
QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating, Electrodeposited, Requirements for

Military

MIL-P-116	Preservation, Methods of
MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (for use with Electronic Equipment in aircraft)
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of

FSC 6615

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Military (Continued)

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-L-6085	Lubricating Oil, Instrument, Aircraft, Low Volatility
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionics Equipment, General Specification for
MIL-C-26482	Connectors, Electric, Circular, Miniature Quick Disconnect
MIL-D-70327	Drawings, Engineering and Associated Lists
MIL-C-27786	Computer Set, Loft Bomb Release AN/AJB-3
MIL-C- 23727	Computer Set, Loft Bomb Release AN/AJB-3A

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS33586	Metals, Definition of Dissimilar
MS17384	Gyroscope, Rate Switching CN-495/AJB-3

2.2 Availability of documents - When requesting any of the applicable documents, refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 9CI, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Preproduction - The rate switching gyroscope furnished under this specification shall be a product, samples of which have been tested and have passed the preproduction tests specified herein.

3.2 Selection of Government documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Commercial parts - Commercial parts having suitable properties shall be used, where on the date of invitations for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, cotter pins, having suitable properties, shall be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and if practicable, on the contractor's drawings.

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3.2.2 Standard parts - With the exception of 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Non-magnetic materials - Non-magnetic materials shall be used for all parts of the rate switching gyroscope except where magnetic materials are essential.

3.3.3 Metals - Metals shall be of the corrosion resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels, salt spray or atmospheric conditions to which the rate switching gyroscope may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings.

3.3.3.2 Magnesium alloy parts - Magnesium alloy parts shall be treated in accordance with Specification MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used.

3.3.3.3 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Small holes, pipe threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.3.3.4 Iron and steel parts - Iron and steel parts which are in hermetically sealed cases shall be cadmium plated in accordance with Specification QQ-P-416. Parts which are not in sealed cases shall be chromium, nickel, or zinc plated in accordance with Specifications QQ-C-320 or QQ-N-290, or QQ-Z-325 respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated.

3.3.4 Protective treatment - When materials are used in the construction of the rate switching gyroscope that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in

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no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.3.5 Lubricants - When required, the lubricant used in the rate switching gyroscope shall be in accordance with Specification MIL-L-6085. The amount of lubricant used shall be sufficient for at least 1500 hours of operation.

3.3.6 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.7 Fumes and vapors - Materials used in the construction of the rate switching gyroscope shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

3.4 Design and construction - The rate switching gyroscope shall be designed to provide switching during turns within specified time intervals at specified turn rates. Design shall be such as to provide unswitching within prescribed time intervals after return to zero turn rate from a particular rate of turn. The rate switching gyroscope shall be constructed to withstand the normal shocks, vibrations and such other conditions as are incident to service without failure, shipping, storage, and installation. The rate switching gyroscope shall operate satisfactorily, when properly connected, in the loft bomb release computer set conforming to the requirements of Specifications MIL-C-23716 or MIL-C-23727.

3.4.1 Maintenance - The design shall be reasonably simple to facilitate as much as possible disassembly, repair or overhaul, service maintenance, and reassembly using those tools and items of maintenance equipment which are normally available as commercial standards.

3.4.2 Case - The rate switching gyroscope case shall conform to the general requirements of Specification MIL-C-172 as specified in this paragraph. The outline dimensions of the case shall conform to MS The case of the gyroscope shall be made of non-ferrous, low density metal, uniform in texture, having a smooth surface and shall be hermetically sealed. The case shall be finished with a lusterless black material, Color No. 37038 of Fed. Std. No. 595.

3.4.3 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein:

3.4.4 Hermetic sealing - The case shall provide a hermetically sealed enclosure for all of the mechanism. The case shall be so constructed that it may be opened, the mechanism removed and replaced and the rate switching gyroscope case resealed at least three times.

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be possible without the use of any special tool, jig or fixture, unless such device is specifically approved by the procuring activity. The sealing of the case shall not be dependent upon any material which will be adversely affected by any atmosphere to which the instrument may be subjected in normal use in military aircraft.

3.4.4.1 Filling medium - The filling medium shall be a mixture of 90 percent nitrogen and ten percent helium. The nitrogen used shall be in accordance with Specification BB-N-411, Type I, Class I, Grade C. The filling medium shall contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 1 ±0.1 atmosphere.

3.4.5 Weight - The rate switching gyroscope shall be so designed that when completely assembled the weight shall not exceed 1.5 pounds.

3.4.6 Screw threads - Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.5 Electrical requirements -

3.5.1 Primary Input Power Requirements - The rate switching Gyro shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power;-Three Phase, 115V category "B", 12VA

3.5.2 Electronic parts - Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on approved lists in accordance with Specification MIL-E-5400 shall not be used unless approved by the procuring activity.

3.5.3 Electronic devices - Electron tubes, transistors and diodes shall be chosen, applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.5.4 Electric connectors - The electric connectors used shall conform to Specification MIL-C-26482 and MS 17382. The connector code number markings as specified in Specification MIL-E-5400 shall be legibly and permanently applied to the case or connector of the rate switching gyroscope.

3.5.5 Wiring - Internal wiring shall be color coded, neat and accomplished in such a manner that individual wires may be easily traced. Wires shall be secured mechanically to the terminals prior to soldering, where practicable. The wiring shall be insulated from the instrument case.

3.6 Performance - The rate switching gyroscope shall perform satisfactorily when subjected to the tests specified in Section 4 of this specification.

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3.6.1 Reliability in mean time between failures - The rate switching gyroscope shall have a minimum of 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.7 Identification of product -

3.7.1 Nameplate - A nameplate conforming to Specification MIL-N-18307 shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130.

3.7.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.8 Interchangeability - For reordered parts, interchangeability shall exist between units and all replaceable assemblies and all sub-assemblies and parts of a designated model of a previously manufactured equipment supplied or designated by the procuring agency for the purpose. Changes in manufactures part number requirements shall be governed by the drawing requirements of MIL-D-70327.

3.9 Installation -

3.9.1 Installation instructions - The contractor shall furnish with each rate switching gyroscope one copy of installation instructions with illustrations and diagrams if necessary. Prior to printing, two copies shall be furnished to the procuring activity for approval. Whenever possible the instructions shall be arranged to require only one sheet of paper either 8-1/2 by 11 or 11 by 17 inches.

3.9.2 Envelope - An envelope furnished by the contractor shall be packaged with each rate switching gyroscope and shall be marked as follows:

IMPORTANT
THIS ENVELOPE CONTAINS
INSTALLATION INSTRUCTIONS

3.10 Workmanship - The rate switching gyroscope, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine-screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

3.10.1 Dimensions - Dimensions and tolerances not specified shall be as close as is consistent with best shop practices. When dimensions and tolerances affect the interchangeability, operation, or performance of the equipment, they shall be held or limited accordingly.

3.10.2 Fabrication - Machining, drilling, and forming shall be done with the use of accurate templates, jigs, or gages.

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3.10.3 Screw assemblies - Assembly screws shall be tight. The word "tight" means that the screws cannot be appreciably tightened without damage or injury to the screw threads or heads.

3.10.4 Riveting - Riveting operations shall be carefully performed in order to insure that the rivets are tight. Rivet heads shall be full size and concentric with the body. Dimensions of the rivets shall be such that the holes will be filled. Loose, cracked, or badly formed, including protruding countersunk rivet heads, shall be replaced before acceptance of the material. Excess metal shall be removed from countersunk rivet heads as a customary operation. This does not cover eyelets or hollow tubular rivets.

3.10.5 Gears - Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. When required for accurate adjustment, gear assemblies shall be free from detrimental backlash.

3.10.6 Cleanup - Units shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material after final assembly. Burrs and sharp edges as well as rosin flash which might crumble shall be removed.

3.10.7 Wire stripping - Wire stripping shall be done without nicking or otherwise damaging the wire.

3.10.8 Soldering - Soldering shall be performed in accordance with Specification MIL-S-6872.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of tests - The inspection and testing of the rate switching gyroscopes shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests performed after award of contract on sample rate switching gyroscopes to determine that the rate switching gyroscope meets the requirements of this specification. The sample rate switching gyroscopes shall be representative in design, performance, and configuration of the rate switching gyroscopes which will be produced on the manufacturer's production line.

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(b) Acceptance tests: Acceptance tests are those tests performed on rate switching gyroscopes manufactured and submitted for acceptance under contract.

4.3 Preproduction tests - The preproduction tests of the rate switching gyroscopes shall consist of all the tests of this specification specified under paragraph headed Test methods.

4.3.1 Preproduction test sample instructions - Preproduction test samples shall consist of three rate switching gyroscopes manufactured in accordance with this specification. The samples submitted for preproduction testing shall not have been previously tested except for the Individual tests. The samples shall be forwarded to the testing laboratory designated by the procuring activity.

4.3.1.1 Preproduction test sample identification - The test samples shall be plainly identified by securely attached, durable tags marked with the following information:

Sample for Preproduction Testing
GYROSCOPE, RATE SWITCHING CN-495/AJB-3
Submitted by (Manufacturer's name, date) for
Preproduction Testing in accordance with
Specification MIL-G-23723 under contract
or order number.
MS 17324
Manufacturer's Part Number

4.4 Acceptance tests - The acceptance tests of the rate switching gyroscope shall consist of the Individual, Sampling and Reliability tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests except that the Sampling plan B tests shall be conducted at a Government laboratory designated by the procuring activity. Acceptance tests, except for Sampling plan B tests, shall be under the supervision of the Government Inspector. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

4.4.1 Individual tests - Each rate switching gyroscope submitted for acceptance shall be subjected to the Individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum each equipment accepted shall have passed the following tests, conducted in the order listed:

Examination of product
Dielectric strength
Switching
Yaw oscillation
Steady state turn and yaw oscillation
Power failure

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4.4.2 Sampling plan tests - The sampling plan tests shall consist of Sampling plan A and Sampling plan B tests. The test samples selected for sampling tests shall first have passed the Individual tests. The test samples which have been subjected to Sampling plan A tests shall not be delivered on contract until they have been refurbished and resubmitted and passed all the Individual tests. Test samples which have been subjected to the Sampling plan B tests shall not be delivered on contract.

4 4.2.1 Sampling plan A sample selection - Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity of Rate Switching Gyroscopes Offered for Acceptance	Quantity to be Selected for Testing
First 15	1 (See *Note)
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

4.4.2.1.1 Sampling plan A tests - Each rate switching gyroscope selected for Sampling plan A testing shall be subjected to the following tests, conducted in the order listed:

Power consumption
Vibration error
High temperature operation
Low temperature operation
Case leakage

4.4.2.2 Sampling plan B instructions - Two rate switching gyroscopes shall be selected at random from the first 15 produced on contract and submitted within 10 days after manufacture. These samples shall be forwarded to a Government Laboratory designated by the procuring activity. Each sample shall be plainly identified by a securely attached, durable tag marked with the following information:

GYROSCOPE, RATE SWITCHING CN-495/AJB-3
Submitted by (Manufacturer's name, date) for
production acceptance Sampling Plan B
testing, in accordance with
Contract/Order No._____.
Manufacturer's part number_____

*NOTE: When Sampling plan B is invoked the quantity shall be zero. When Sampling plan B is to be omitted the quantity shall be one.

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4.4.2.2.1 Sampling plan B tests - Each rate switching gyroscope selected for Sampling plan B testing shall be subjected to the following tests, conducted in the order listed:

Sampling plan A
Radio noise interference
Temperature-altitude
Vibration failure
Shock
Life
Test for helium
Internal examination

4.4.3 Reliability assurance tests - The test samples which have successfully passed the Individual tests shall be tested for reliability quality assurance in accordance with Specification MIL-R-23094 Procedure I.

4.4.3.1 Test level - The test level shall be as specified in the system Specification MIL-C-23716 or MIL-C-23727.

4.4.3.2 Duty cycle - The duty cycle shall be as specified in the system Specification MIL-C-23716 or MIL-C-23727 .

4.4.3.3 Accept-Reject criteria - Figure 13 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Qualification plan and figure 14 shall be used for the Reliability Sampling plan.

4.4.3.4 Performance characteristics to be measured - The performance of the rate switching gyroscope shall not degrade the system to the extent defined as a failure in the "Performance characteristics to be measured" paragraph of the system Specification MIL-C-23716 or MIL-C-23727.

4.4.3.5 Failure criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(a) Whenever performance characteristics fall below the acceptance requirements (4.4.3.4) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure, unless one part caused the other parts to fail.

(b) If more than one item is responsible for degrading the system each item shall be counted as a failure.

4.4.3.6 Preventive maintenance - During the test no preventive maintenance, other than simple adjustments normally performed on the equipment by the operator, shall be permitted. However, a log of all such adjustments shall be kept. The procuring activity may weigh these adjustments and if it feels justified assign excessive ones as failures.

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4.4.3.7 Additional requirements - The procedure set forth in 4.4.3.1 requires a complete system, or suitable simulation where the test is applied for item reliability requirements for the item specified herein. System and item reliability figures may be obtained in simultaneous tests if calculation is suitably apportioned.

4.5 Test conditions -

4.5.1 Standard conditions - Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient 25°C ±5C
Pressure	Normal atmospheric (approximately 29.92 inches Hg.)
Humidity	Room ambient up to 90% relative humidity

4.5.2 Attitude - The rate switching gyroscope shall be secured with mounting feet down to a rate table, capable of rates up to 180° per minute.

4.5.3 Input power - The rate switching gyroscope shall be tested with 115 ±2.3 Vac, single phase, 400 ±5 cps applied between pins A and B (neutral) of J101.

4.5.4 Test equipment and connections - Special test equipment and connections used in performing the tests shall not affect the performance of the rate switching gyroscope under test. If the performance is affected by the test equipment, proper compensation shall be made.

4.6 Test methods -

4.6.1 Examination of product - Each rate switching gyroscope shall be examined externally to determine conformance with the applicable drawings and with all the requirements of this specification not covered by tests.

4.6.2 Dielectric strength - A potential of 500 Vdc shall be applied between isolated circuits and between each isolated circuit and the case or connector shell for a period of 10 seconds. The insulation resistance shall be not less than 50 megohms.

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4.6.3 Switching -

4.6.3.1 Secure the rate switching gyroscope, with mounting feet down, to the surface of a rate table. Apply power as specified in 4.5.3. Allow the rate switching gyroscope to run for one minute minimum before starting test. All switching tests shall be performed in the sequence of the highest specified turn rate applied first and then on through the lowest specified turn rate.

4.6.3.2 Adjust the rate table for 180° per minute rate of turn, then start rotation in a clockwise (CW) direction. Continuity between pins D and E, and an open circuit between pins C and D, shall occur within 3 seconds.

4.6.3.3 Stop rotation; an open circuit between pins D and E, and continuity between pins C and D, shall occur within 20 seconds after rotation stops.

4.6.3.4 Repeat the procedure as specified in 4.6.3.2 and 4.6.3.3 at turn rates of 45°, 20°, and 15° per minute respectively. Switching shall occur at each of these turn rates within 8 seconds, 20 seconds, and 30 seconds, respectively. Unswitching shall occur as specified in 4.6.3.3 after stop of each specified rate of turn.

4.6.3.5 Repeat the procedure as specified in 4.6.3.1, 4.6.3.2, 4.6.3.3, and 4.6.3.4 except the direction of rotation shall be in the counterclockwise (CCW) direction. The same results shall be obtained.

4.6.4 Yaw oscillation - Secure the rate switching gyroscope with mounting feet down on a sinusoidal rate table and apply power as specified in 4.5.3. Allow the rate switching gyroscope to run for one minute minimum before beginning test. At zero turn rate, apply yaw oscillations of plus and minus one degree per second with a 2 second period; the open circuit shall remain between pins D and E, and continuity shall continue to exist between pins C and D.

4.6.5 Steady state turn and yaw oscillation - Secure the rate switching gyroscope, with mounting feet down, to a combined rate and sinusoidal rate test apparatus and apply power as specified in 4.5.3. Allow the rate switching gyroscope to run for one minute before beginning test. Superimpose yaw oscillations of plus and minus one degree with a 2 second period on a 30° per minute rate of turn in the CW direction. Continuity between pins D and E, and an open circuit between pins C and D shall occur within 30 seconds after start of turn. Repeat this procedure except in CCW direction of rotation; the same results shall be obtained.

4.6.6 Power failure - Secure the rate switching gyroscope, with mounting feet down to a rate table and apply power as specified in 4.5.3. Allow the rate switching gyroscope to run for one minute before beginning test. Rotate the rate switching gyroscope at 20° per minute turn rate. After switching has occurred, remove power from the rate switching gyroscope. An open circuit between pins D and E, and continuity between pins C and D, shall occur immediately upon removal of power. Repeat this procedure except in a CCW direction of rotation, the same results shall be obtained.

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4.6.7 Power consumption - Apply power as specified in 4.5.3. The power consumed by the rate switching gyroscope shall not exceed 10 volt-amperes.

4.6.8 Vibration error - The vibration error test shall consist of a frequency survey with vibration applied along the three principal axes of the rate switching gyroscope. The rate switching gyroscope shall be subjected to vibration with a constant applied double amplitude of 0.060 inch through the frequency range of 10 to 55 cps. The duration of the vibration shall be 15 minutes in each axis. Satisfactory performance shall be determined by observing that continuity does not appear between pins D and E during vibration.

4.6.9 High Temperature Operation - The rate switching gyro shall be properly connected and power applied as in 3.5.1. The rate switching gyro shall then be subjected to an ambient temperature of 71.°C (160°F) for a period of 4 hours. At the end of the four hour period, the unit shall be subjected to a temperature of 95°C for a period of 30 minutes and then shall meet the requirements of the Sampling Plan A Tests.

4.6.10 Low temperature operation - The rate switching gyroscope shall be properly connected except that no power shall be applied. The rate switching gyroscope shall then be subjected to an ambient temperature of -54°C (-65°F) for a period of 4 hours. Power shall be applied as specified in 4.5.3 during the last 15 minutes of the 4-hour period. With the ambient temperature maintained at -54°C (-65°F), the rate switching gyroscope shall then meet the requirements as specified in 4.6.3, excepting that the time to switch at 180°, 45°, 20°, and 15° rate of turn shall not exceed 10, 20, 45, and 65 seconds respectively. Time to unswitch (see 4.6.3.3) after return to zero rate of turn from each of these rates shall not exceed 100 seconds. The rate switching gyroscope may be gently vibrated or tapped during this operation check.

4.6.11 Case leakage - Each rate switching gyroscope shall be tested for case leakage with either a mass spectrometer or by an equally sensitive leak test. The gas or mixture of gases in the case, adjusted to a pressure differential of one atmosphere, shall not have a leakage rate from the case exceeding one micron cubic foot per hour (10^{-5} cubic centimeters per second).

4.6.12 Radio noise interference - The electrical system of the rate switching gyroscope shall be tested for conducted and radiated radio noise interference and susceptibility, in accordance with Specification MIL-I-6181 over a frequency range not exceeding 1000 mc. Permissible deviation from short duration interference limits as specified in Specification MIL-I-6181 shall apply as necessary.

4.6.12.1 Conducted interference - Power as specified in 4.5.3 shall be applied during this test. A rate of turn, sufficient to cause the rate switching gyroscope to switch, shall be applied a minimum of 3 times per frequency octave.

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4.6.12.2 Radiated interference - Conditions as specified in 4.6.12.1 shall be applied during this test.

4.6.12.3 Susceptibility - The rate switching gyroscope shall meet the requirements as specified in 4.6.4 during conducted susceptibilty powerline and radio frequency radiated tests.

4.6.13 Temperature-altitude - The rate switching gyroscope shall be subjected to the temperature-altitude test of Specification MIL-E-5272, Procedure I, subject to the exceptions provided for equipment designed for continuous operation at 71°C (160°F).

(a) A check of performance shall be made during steps 2 and 4 of Procedure I by applying power as specified in 4.5.3 for 15 minutes (after temperature has stabilized); then subjecting the rate switching gyroscope to the test as specified in 4.6.3, except the switching and unswitching times as specified in 4.6.10 shall apply. The rate switching gyroscope may be gently vibrated or tapped during this check.

(b) During each of steps 7, 8, and 10 of Procedure I, the rate switching gyroscope shall be subjected to the test as specified in 4.6.3 once each hour of the specified 4 hour period, except that the unswitching time as specified in 4.6.9 shall apply.

4.6.14 Vibration Failure - The equipment shall be tested in accordance with Procedure XII of Specification MIL-E-5272 except the frequency shall be 5 to 2000 cps.

4.6.15 Shock - The rate switching gyroscope shall be subjected to the shock test of Specification MIL-E-5272, Procedure V. At the end of this test the rate switching gyroscope shall meet the requirements of the Individual tests.

4.6.16 Humidity - The rate switching gyroscope shall be subjected to the humidity test of Specification MIL-E-5272, Procedure I.

4.6.17 Fungus resistance - The rate switching gyroscope shall be subjected to the fungus resistance test of Specification MIL-E-5272, Procedure I, except that the test period shall be 14 days.

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4.6.18 Salt spray - The rate switching gyroscope shall be subjected to the salt spray test of Specification MIL-E-5272, Procedure I.

4.6.19 Life - Secure the rate switching gyroscope with the mounting feet down, to a Scorsby table. Apply power as specified in 4.5.3. Subject the rate switching gyroscope to a 15° total movement at 5 to 7 oscillations per minute. Power and movement shall be applied for 22 hours, then shut off for 2 hours. This cycling shall be continued until a total of 1500 hours operating time has been reached. The rate switching gyroscope shall meet the requirements of the Individual tests prior to every 250 hours and at the completion of 1500 hours of operation.

4.6.20 Test for helium - The rate switching gyroscope case shall be punctured, or the filling tube cut, and the rate switching gyroscope subjected to the test for helium by means of a mass spectrometer type helium detector. Failure to detect helium shall be cause for rejection.

4.6.21 Internal examination - The case of the rate switching gyroscope shall be opened and the internal mechanism examined. Any deterioration or damage which could in any manner prevent the rate switching gyroscope from meeting functional operation or maintenance requirements during service life shall be cause for rejection.

4.7 Rejection and Retest - Signal adapters which have been rejected as a result of individual tests may be reworked and resubmitted for acceptance. Failure of adapters during either the Sampling Plan A or B tests shall terminate acceptance of adapters until the cause of the failures have been determined and satisfactory remedial actions agreed upon by the procuring agency.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The rate switching gyroscopes shall be packaged in accordance with Specification MIL-P-7936, Level A. The method of preservation shall be in accordance with Specification MIL-P-116, Method 1A-5 without preservative compound, using metal reusable containers.

5.2 Packing - The rate switching gyroscopes shall be packed in accordance with Specification MIL-P-7936, Level A, B or C as specified in contract or order (see 6.2).

5.3 Marking - The interior and exterior containers shall be marked as specified in Specification MIL-P-7936.

5.3.1 Precautionary marking - The following precautionary markings shall appear on two opposite sides of each interior package and shipping container whenever practicable.

FRAGILE
DELICATE INSTRUMENT
HANDLE WITH CARE

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6. NOTES

6.1 Intended use - The rate switching gyroscope covered by this specification is intended for use in aircraft as a unit of the AN/AJB-3 or AN/AJB-3A loft bomb release computer set.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, Number and Date of this specification.

(b) The quantity, MS part number and type number, if any, of the instrument desired.

(c) Level of Packing Desired (See 5.2)

6.3 Drawing requirements - The attention of contractors is invited to the fact that Government contracts for equipment, whether procured directly by the Government or through another contractor, require that all drawings submitted must be in accordance with the requirements of Specification MIL-D-70327.

6.4 Definitions -

6.4.1 Amplitude - Whenever the word "amplitude" is specified, it shall mean the extent of motion as measured from one extreme to the opposite extreme.

6.4.2 Hermetic seal - A hermetic seal shall be a perfectly closed and air tight seal made between metallic and metallic or between metallic and vitric materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.5 Preproduction test provisions - The manufacture of rate switching gyrocopes on contract shall not commence until the samples submitted are pronounced satisfactory by the procuring activity. When a contractor is in continuous production of these rate switching gyroscopes from contract to contract, the submission of further preproduction samples on the subsequent contracts may be waived at the discretion of the procuring activity. Approval of preproduction samples or the waiving of preproduction tests does not preclude the requirement of acceptance testing.

6.5.1 It shall be understood that the rate switching gyroscopes supplied under contract or order shall be identical to the corresponding preproduction sample in design, construction, quality, material, workmanship, and method of manufacture. Deviation from the standards of the preproduction sample shall be made only by the procuring activity. Evidence of unauthorized change shall constitute cause for rejection.

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NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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DEPARTMENT OF THE NAVY
BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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NAVY DEPARTMENT

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/8 (8-62)

FORM APPROVED
BUDGET BUREAU NO 45-R300

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No and abbrev title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1 HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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